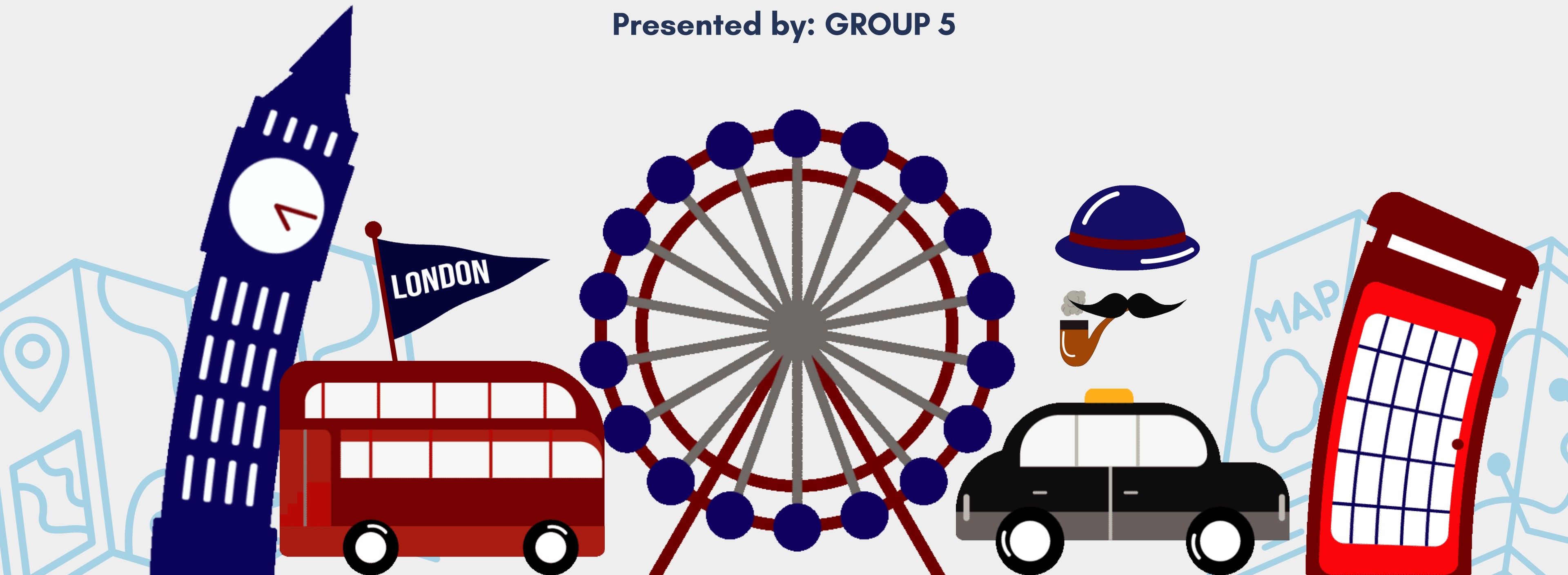


LONDON AIRBNB

DATA-230 SEC-11 DATA VISUALIZATION

Presented by: GROUP 5



MEET OUR LONDON DATA CREW



Priya



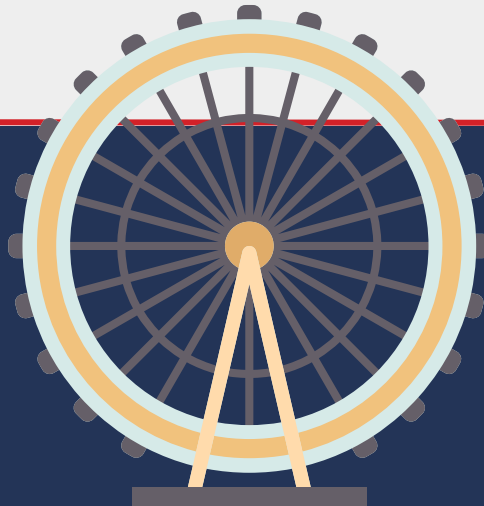
Varuna



Sanjana



Ngoc



Research Question:

WHAT MAKES A LONDON AIRBNB TOP-RATED - 4.8+?

And what can hosts learn from the listings that get it right?



Our goal: Turn 70,708 listings into actionable insights for London Airbnb hosts

Analysis journey

OUR ROADMAP TO THE ANSWER



**THE
MARKET**

**What does
London Airbnb
look like?**

**THE
HOST**

**Does the host
behind the
listing matter?**

**THE
LOCATION**

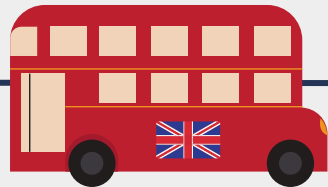
**Where are
highly rated
listings
concentrated?**

**THE
LISTING**

**Which property
and listing
features move
together with
ratings?**

**THE
ANSWER**

**What should a
host focus on?**

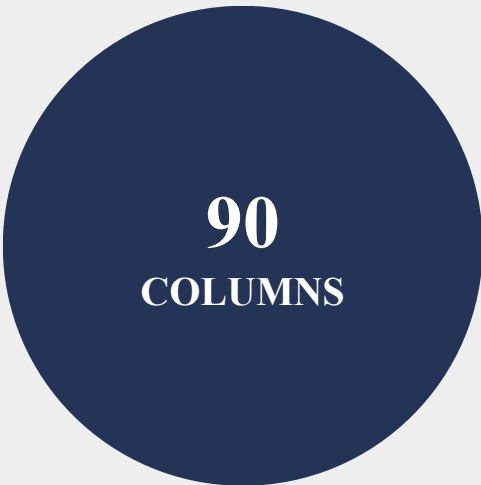


Dataset overview

92,638 LISTINGS: WHAT EARNS A 4.8+?



One row = one listing



Price, location, property, host, reviews



of 70,708 guest-rated listings scored 4.8+

SOURCE
Inside Airbnb, London
Snapshot 19 June 2026 · listings.csv.gz · prices in GBP
insideairbnb.com/get-the-data

RAW RECORDS: 3 EXAMPLE ROWS

price	bathrooms_text	bedrooms
\$127.00	1 shared bath	NaN
\$108.00	1 shared bath	1
NaN	1 bath	2

OUR TASK
Predict whether a listing is top rated (4.8+) so hosts know what to improve first.

WHY THIS DATASET
UCI Restaurant: 7,207 rows, too small
Open Food Facts: 2017, computed target
Airbnb London: 92,638 rows, 2026, guest ratings

WHY IT'S MESSY
14 empty columns
Price stored as text
33% of prices missing
Hidden placeholders
24% of listings unrated

Data cleaning

4 DECISIONS, EACH WITH A REASON

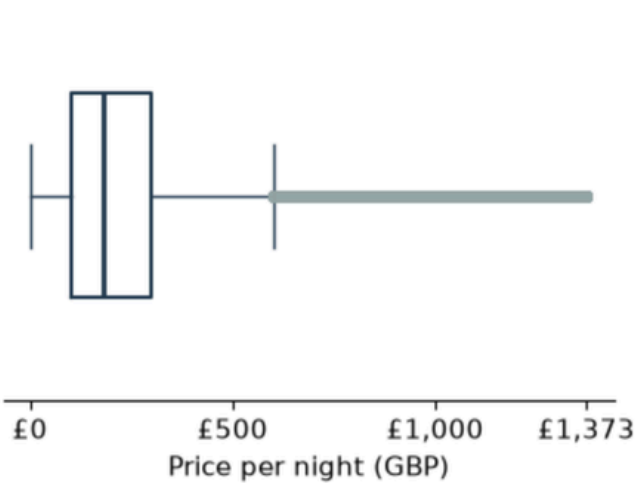
	1 DROP	2 CONVERT	3 CAP	4 FILL + FLAG
ISSUE	14 columns 100% empty	Price stored as text ("£127.00")	£527,524 price; rating 0; 2,147,483,647 nights	Missing price, beds, bedrooms, bathrooms
DECISION	Dropped	Converted to a number	Capped at £1,373 (new column); placeholders → missing	Room-type median + "missing" flag
BECAUSE	No values in this snapshot	Text cannot be charted or modelled	Entry errors and placeholders, not real values	Skewed values; missingness linked to rating

BEFORE AND AFTER

Before: listed price



After: capped at £1,373 (99th percentile)

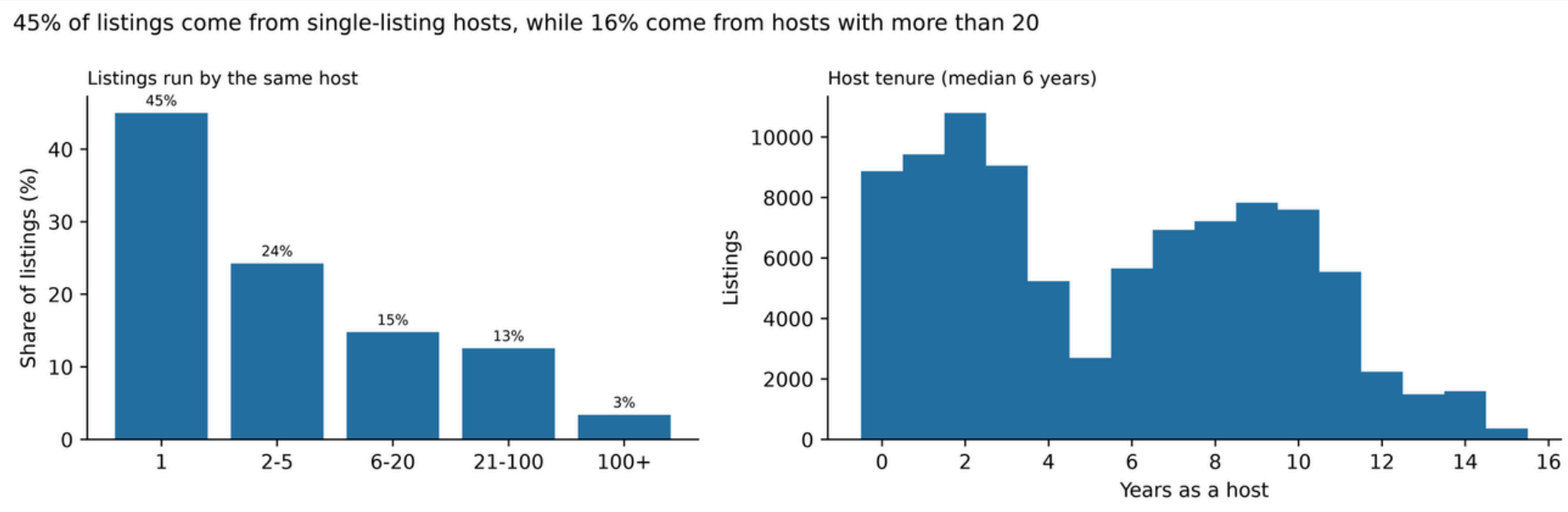


THE OUTLIER £527,524
One night, an entry error. 623 prices capped; no rated listing deleted.

CHALLENGE
33% of prices missing; 87% of those can't be booked: a pattern, not random.
Result: 92,635 × 80 · 7/7 checks pass

Univariate overview : Host Profile

WHO IS BEHIND LONDON'S AIRBNBS?



TAKEAWAY

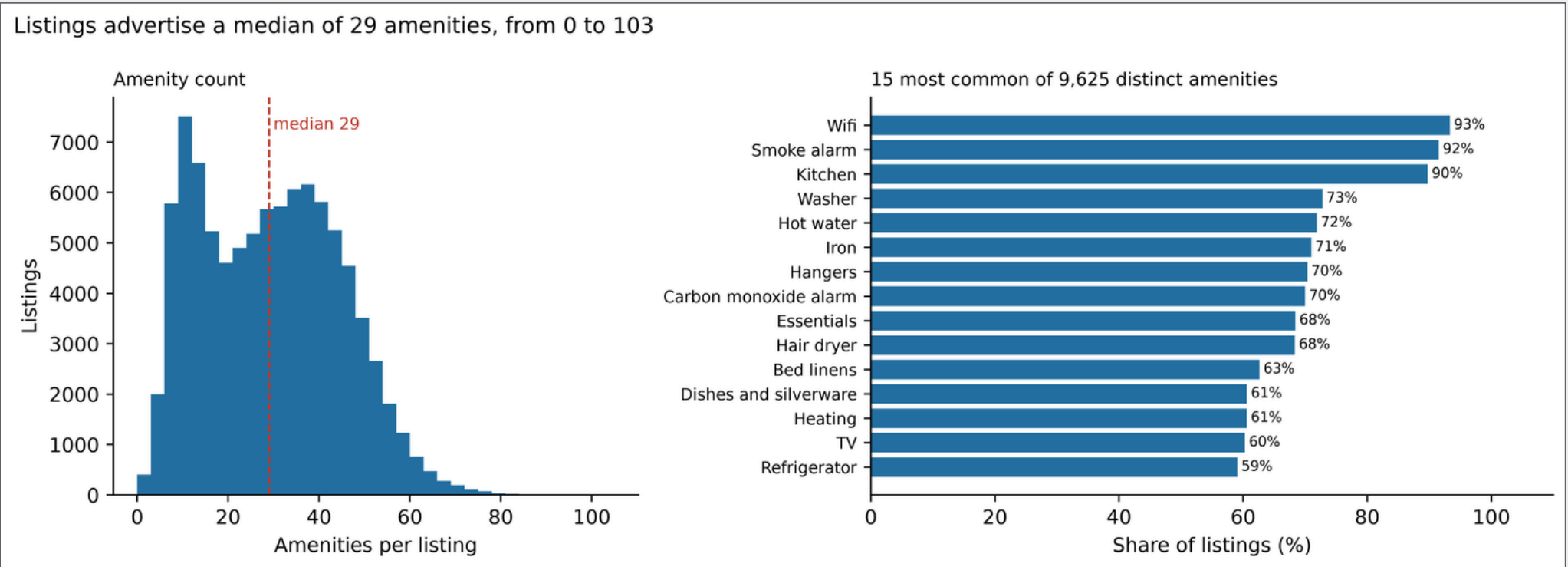
London’s host landscape is diverse: from single-listing hosts to large-scale operators.

This makes portfolio size + host tenure worth testing against ratings.

OBSERVATION	DECISION
45% from single-listing hosts; portfolios vary widely	Test portfolio size → ratings
Median tenure = 6 years	Test host tenure → ratings

Univariate overview: Listing Amenities

WHAT DO LONDON AIRBNBS OFFER?



TAKEAWAY

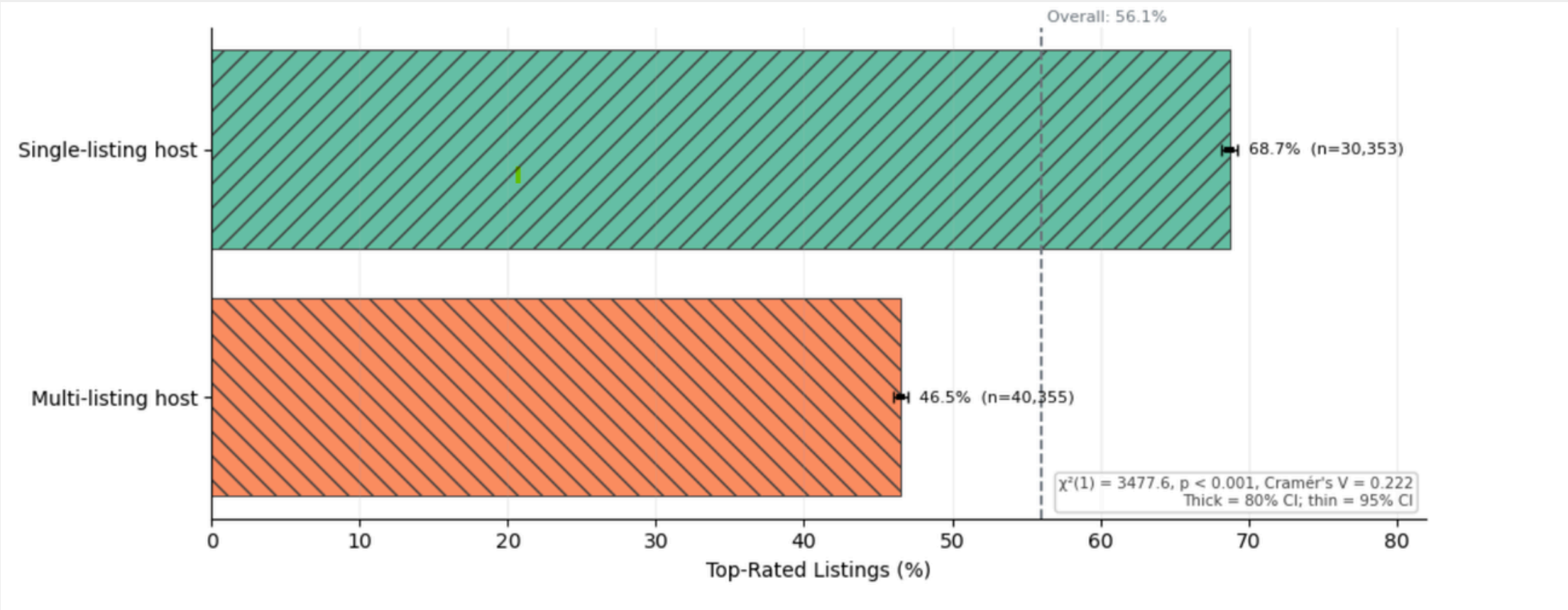
London listings vary substantially in amenity count, while basic amenities are nearly universal.

Amenity quantity and differentiating amenities move forward to rating analysis.

OBSERVATION	DECISION
Median = 29 amenities, with wide variation across listings	Test amenity count → ratings
WiFi (93%), smoke alarm (92%) & kitchen (90%) are nearly universal	Focus later analysis on amenities that differentiate listings

Bivariate overview: Host Profile

DOES HOSTING MORE LISTINGS MEAN BETTER RATINGS?



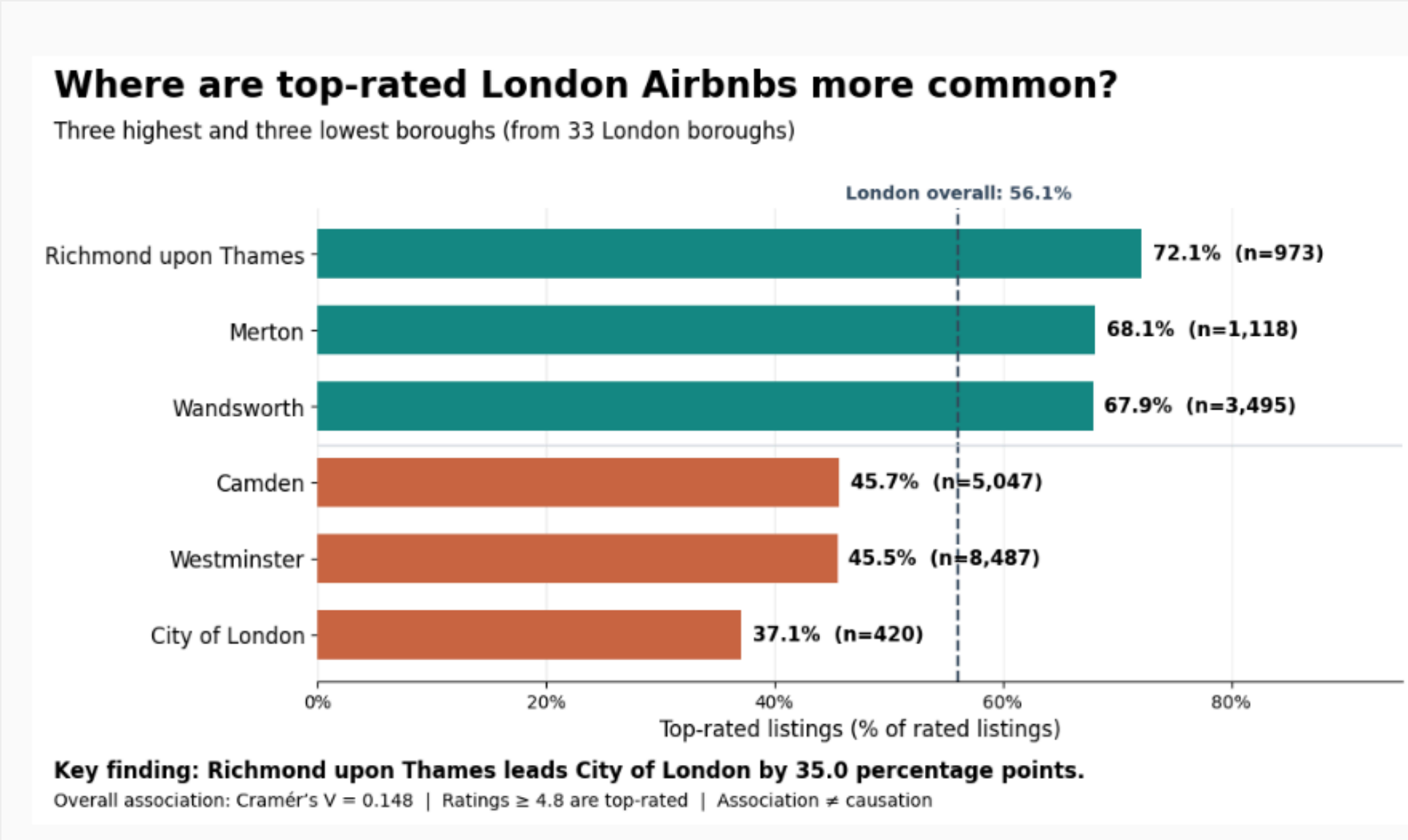
TAKEAWAY

Single-listing hosts show a 22.2 percentage-point higher top-rated rate. Host type is our strongest categorical bivariate signal.

OBSERVATION	DECISION
Top-rated: 68.7% single-listing vs 46.5% multi-listing hosts	Retain host type as a high-priority feature
Cramér's V = 0.222, strongest categorical association examined	Carry host portfolio size into multivariate analysis

Bivariate overview: Location

DOES LOCATION INFLUENCE THE GUEST EXPERIENCE?



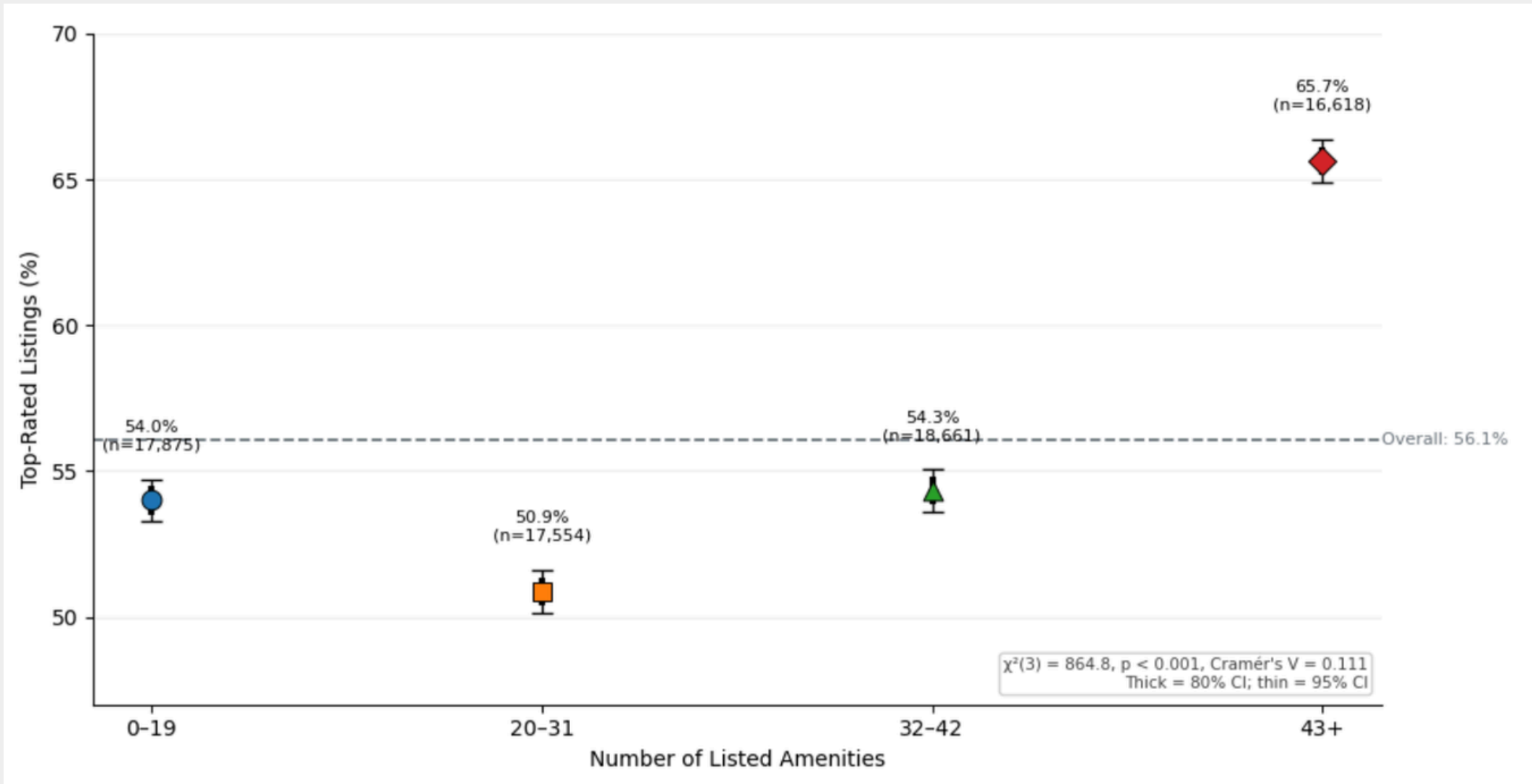
TAKEAWAY

Top-rated performance varies widely across London boroughs.
Location matters, but host type remains the stronger signal.

OBSERVATION	DECISION
Top-rated rates range from 72.1% in Richmond upon Thames to 37.1% in City of London	Retain borough/location for multivariate analysis
Cramér's V = 0.148 meaningful but weaker than host type (0.222)	Treat location as a supporting predictor, not the strongest signal

Bivariate overview: Amenity

WHICH LISTING FEATURES MOVE WITH RATINGS?



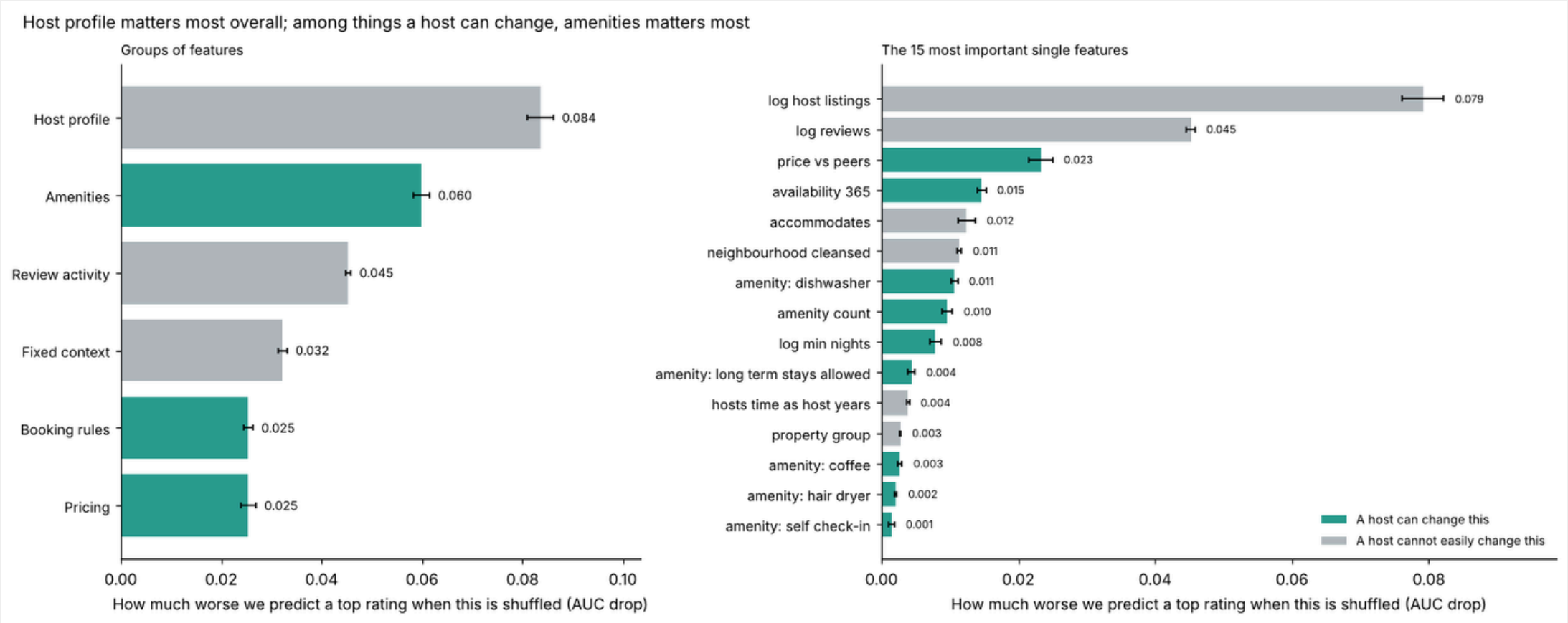
TAKEAWAY

Amenity-rich listings (43+) show the highest top-rated rate at 65.7%. Amenities matter, but their signal is weaker than host type and location.

OBSERVATION	DECISION
Listings with 43+ amenities reach 65.7% top-rated, highest among the groups	Retain amenity count for multivariate analysis
Cramér's V = 0.111 association exists, but is weaker than host type & location	Treat amenities as a supporting feature, not a dominant driver

Multivariate overview: Combined Signals

WHAT HAPPENS WHEN WE CONNECT THE SIGNALS?



TAKEAWAY

No single feature predicts a top rating (none above $p = 0.28$), but together they reach AUC 0.770.

Host profile matters most, while amenities are the strongest factor hosts can change.

OBSERVATION	DECISION
Host profile has the largest grouped importance (0.084)	Keep host profile as key context
Amenities lead host-controllable factors (0.060)	Prioritize amenities in host recommendations
Host-type gap persists in 32 of 33 boroughs	Keep host type + location together

Preliminary ML direction

CAN WE PREDICT WHAT DRIVES SUCCESS?

TARGET	INPUT SIGNALS	MODEL PLAN
Top rated ≥ 4.8 (1)	Host profile	Logistic regression (baseline)
Not top rated < 4.8 (0)	Location (borough)	Random forest
70,708 rated listings	Listing + amenities	Compare on validation set
56.1% top rated	Booking rules + pricing	Final check on test set

ML TASK: BINARY CLASSIFICATION

Why, from our EDA:

Ratings pile up near 5.0, so we predict top rated: yes or no

56/44 balance, so no resampling needed

Combined features beat a straight-line rule (AUC 0.770 vs 0.725), so we compare with a tree model

Review sub-scores and superhost excluded (target leakage)

MODELING DECISION

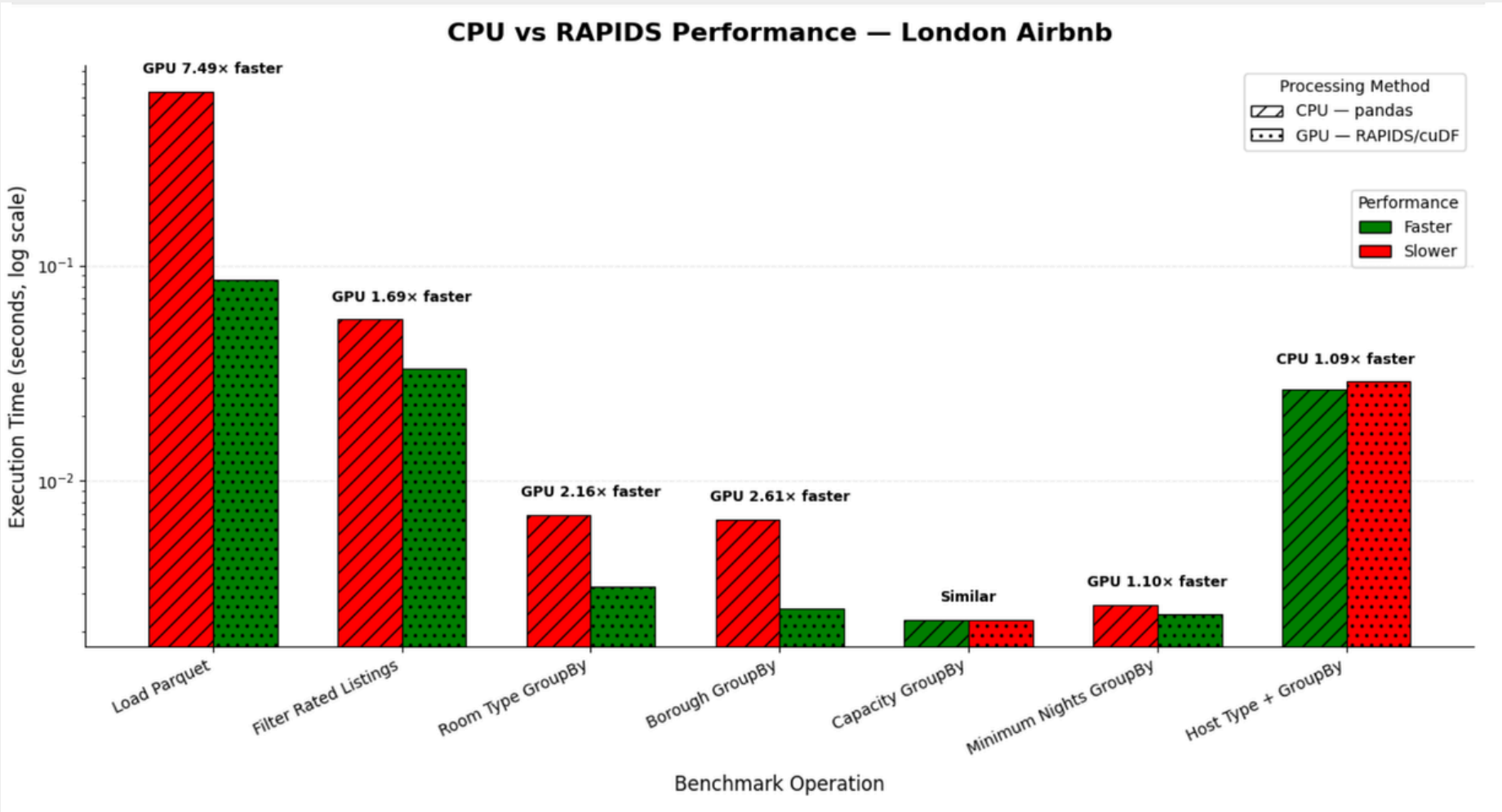
Stratified 70/15/15 split | Preprocessing fit on training rows only | Compare F1, ROC-AUC and confusion matrix

TAKEAWAY

EDA identified the signals; ML tests how well they predict top-rated status together.

CPU vs GPU (RAPIDS)

DOES GPU ACCELERATION ACTUALLY HELP?



Operation	CPU (seconds)	GPU (seconds)
Data Loading (Parquet)	0.6411	0.0856
Filtering Rated Listings	0.0562	0.0332
Room Type GroupBy	0.0070	0.0032
Borough GroupBy	0.0066	0.0025
Capacity GroupBy	0.0023	0.0023
Minimum Nights GroupBy	0.0027	0.0024
Host Type + GroupBy	0.0266	0.0289

DECISION

Use RAPIDS for data loading and heavier aggregations; retain pandas for lightweight operations.

TAKEAWAY

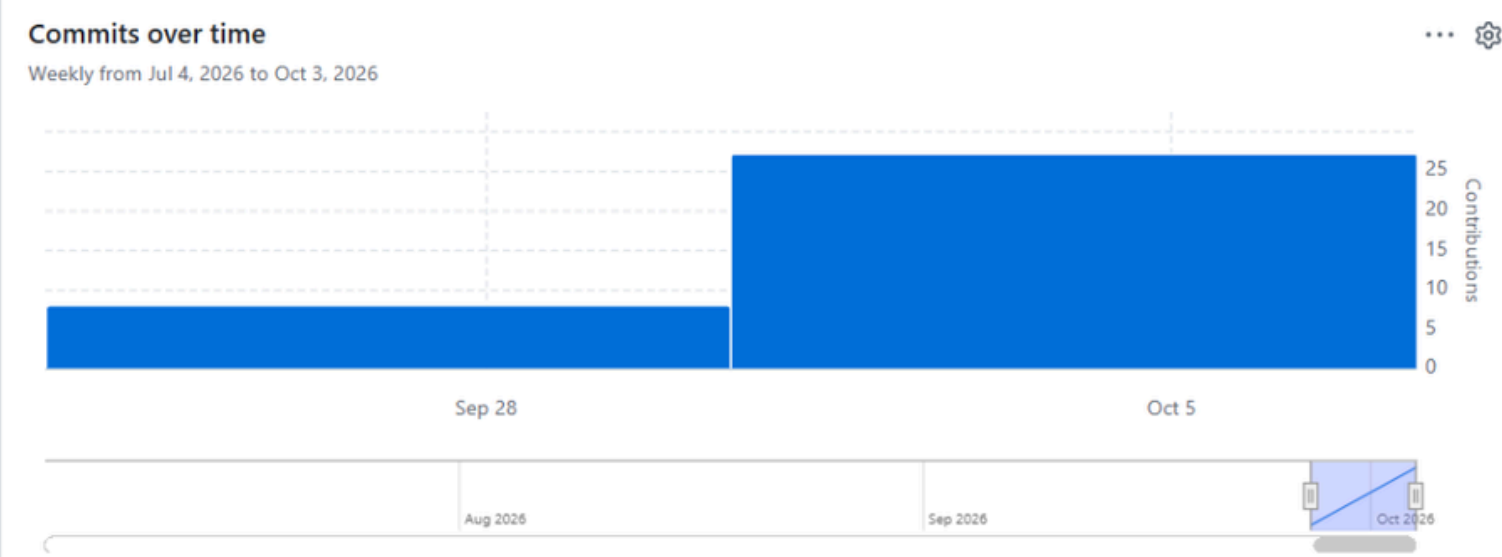
GPU ≠ always faster. RAPIDS delivers the biggest gains on heavier workloads, while pandas remains competitive on smaller operations.

Team contribution

WHO BUILT WHAT

GITHUB LINK

MEMBER	OWNED
Varuna	Dataset selection, cleaning, CPU timing
Ngoc	Univariate analysis
Sanjana	Bivariate, ML direction, CPU vs GPU
Priya	Multivariate
Priya	Tableau dashboard



Final synthesis

SO, WHAT MAKES A TOP-RATED AIRBNB?

HOST MATTERS MOST	LOCATION MATTERS	AMENITIES HELP
68.7% top-rated for single-listing hosts vs 46.5% for multi-listing hosts	Top-rated rates vary from 37.1% 72.1% across boroughs	Listings with 43+ amenities reach 65.7% top-rated
Strongest categorical signal $V = 0.222$	Supporting signal $V = 0.148$	Actionable signal $V = 0.111$

FINAL MESSAGE:

Top ratings are not driven by one feature. Host profile provides the strongest overall signal, while amenities are the strongest factor hosts can directly change.

Hosts cannot change their location or history but they can improve how their listing competes within that context.



70,708 London Airbnbs
One question. One clear takeaway.

**Top-rated success comes from a
combination of context and choices,
amenities are the strongest lever hosts
can directly control**

THANK YOU